Structural Biology Report

Modelling L-Lactate Dehydrogenase Complexes

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Course: Structural Bioinformatics
Date of completion: 3 of March - 2026

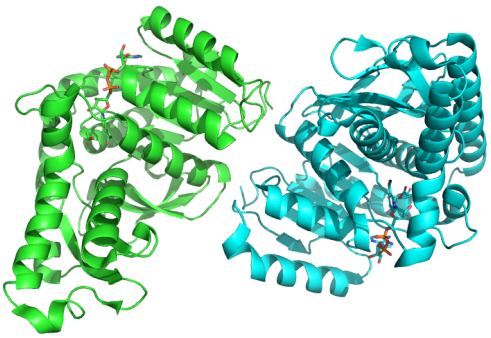
Introduction:

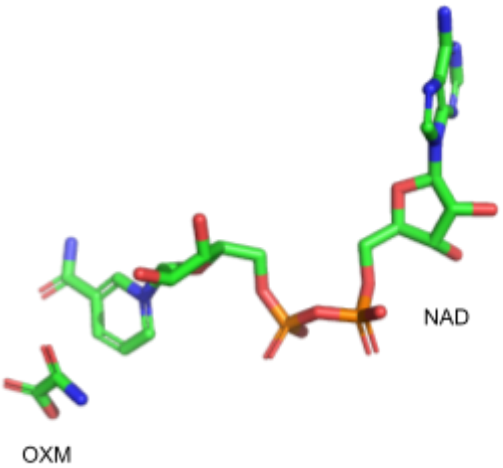
The goal of this practice is to model the allosteric activation of L-Lactate dehydrogenase (LDH) from B. *Stearothermophilus.* We focused on building different catalytic complexes using PDB structure **1LDN** as our starting point to understand the interaction between the enzyme, its substrates (pyruvate and NADH), and its activator (FBP).

Modeling the LDH with substrates (Pyruvate, NADH and H^+)

Step1: System Cleaning and Dimer Selection

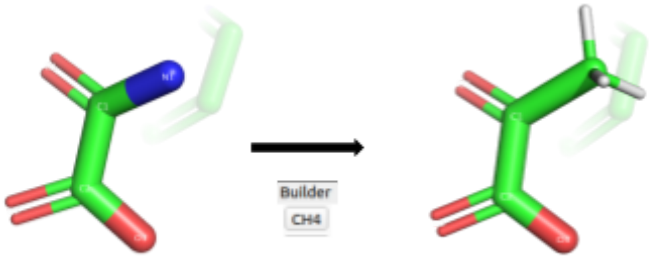
We began by fetching the 11dn PDB structure, which contains two tetramers. We created a selection named LDH_dimer using only chains A and B. To clean the system, we removed all water molecules and kept only the protein polymer and the essential ligands: NAD and Oxamate (OXM)





Step 2: Converting Oxamate to Pyruvate

Following the reaction we are modelling, L-Lactate dehydrogenase catalyzes the conversion of Pyruvate to Lactate. Since Oxamate is a substrate analogue, we converted it into Pyruvate by replacing its amino group (NH_2) with a methyl group (CH_3). We identified and removed the nitrogen atom in OXM and used the Builder tool to attach a methyl group to carbon C1. While the tool displays the fragment as CH_4, it becomes the correct CH_3 group once bonded to the substrate structure

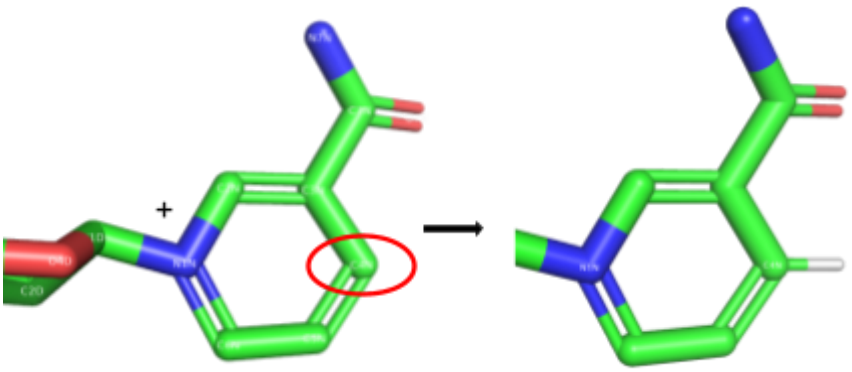


Step 3: Conversion of NAD to NADH

To complete the catalytic machinery, we simulated the reduction of the cofactor:

$$NAD^{+} + H^{-} \rightarrow NADH$$

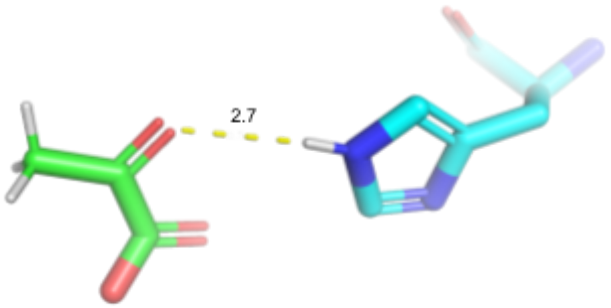
We added a hydride to the C4 position of the nicotinamide ring, which is opposite to the nitrogen atom. Although PyMOL does not explicitly model formal charges , the resulting arrangement of the ring after the hydrogen addition properly represents the NADH molecule.



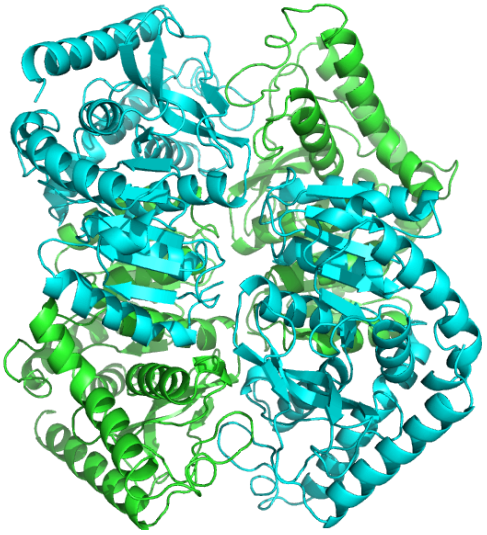
Step 4: Protonation of the Catalytic Residue (H^+)

Finally, we modeled the source of the proton needed to form the hydroxyl group in lactate. When the hydride attacks the carbonyl carbon of pyruvate, the resulting negatively charged oxygen must be protonated by a catalytic residue.

We identified **Histidine 193** (His193) in both chains A and B as the proton donor. Using the builder, we attached a hydrogen to the histidine and measured a distance of **2.7 Å** to the substrate to ensure the interaction is plausible.



Building the Tretamer with FBP Activator



Reconstruction the Tetramer

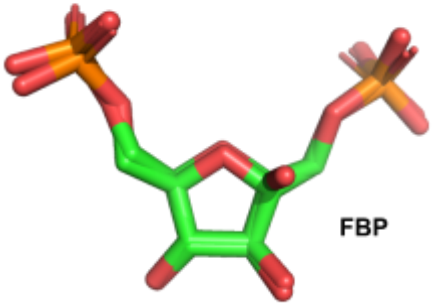
To study the allosteric activation, we moved from the dimer model to the full tetramer structure.

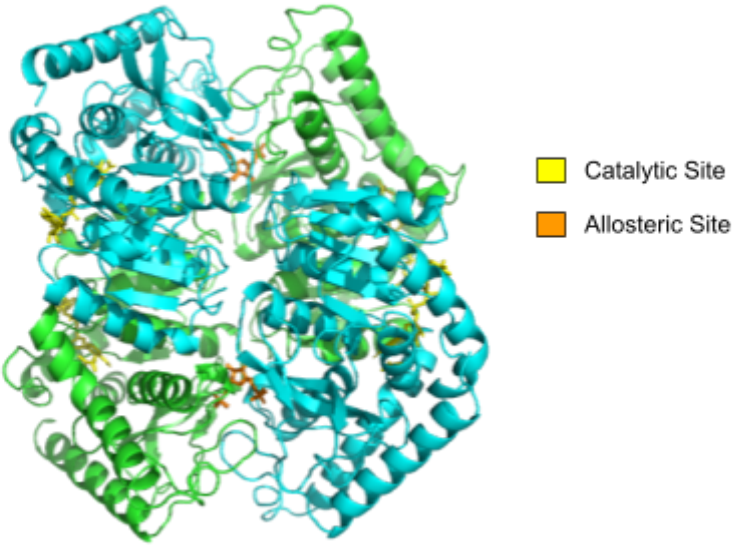
We fetched the **1LDN** PDB again, but this time we kept all four chains (A, B, C, and D) to represent the complete functional enzyme unit.

Identifying the Allosteric Sites

L-Lactate dehydrogenase from B. Stearothermophilus is regulated by **Fructose - 1,6 - BisPhosphate (FBP).** We identified the two regulatory sites located at the interfaces between the subunits where the activator binds to trigger the conformational change.

Positioning the FBP Molecules



Using the **Bilder toll** and coordinates from known regulatory complexes, we placed two molecules of FBP into the tetramer.

Each FBP molecules interacts with specific residues from different chains, stabilizing the active state of the enzyme.